

Taylor's Series Expansions / Taylor's Thm

① Expansion of $f(x)$ in terms of $(x-a)$ is.

$$f(x) = \underline{f(a)} + \underline{(x-a)} \underline{f'(a)} + \frac{(x-a)^2}{2!} \underline{f''(a)} + \frac{(x-a)^3}{3!} \underline{f'''(a)} + \dots$$

(constant term)

② Expansion of $f(x+h)$ in powers of h

$$f(x+h) = f(x) + \underline{h} f'(x) + \frac{h^2}{2!} f''(x) + \frac{h^3}{3!} f'''(x) + \dots$$

③ Expansion of $f(x+h)$ in powers of x .

$$f(x+h) = f(h) + \underline{x} f'(h) + \frac{x^2}{2!} f''(h) + \frac{x^3}{3!} f'''(h) + \dots$$

★ Maclaurin Series:

It is a Taylor's series which we'll get by subst. $a=0$ in ①

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$$

$$e^x = 1 + x + \frac{x^2}{2!} + \dots$$

① Expand e^x in powers of $(x-1)$

$$\left(\frac{(x-2)}{\frac{(x-3)}{(x-1)}} \right)$$

By Taylor's series

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \dots$$

$$a=1$$

$$f(x) = e^x = f(1) + (x-1)f'(1) + \frac{(x-1)^2}{2!}f''(1) + \dots \quad \textcircled{1}$$

$$f(x) = e^x$$

$$f(1) = e$$

$$f'(x) = e^x$$

$$f'(1) = e$$

$$f''(x) = e^x$$

$$f''(1) = e$$

from ①

$$e^x = e + (x-1)e + \frac{(x-1)^2}{2!}e + \frac{(x-1)^3}{3!}e + \dots$$
$$= e \left[1 + (x-1) + \frac{(x-1)^2}{2!} + \frac{(x-1)^3}{3!} + \dots \right]$$

② Expand $\tan^{-1}x$ in powers of $(x - \frac{\pi}{4})$

$$f(x) = f(a) + (x-a) f'(a) + \frac{(x-a)^2}{2!} f''(a) + \dots$$

$$a = \frac{\pi}{4}$$

$$f(x) = \tan^{-1}x = f\left(\frac{\pi}{4}\right) + \left(x - \frac{\pi}{4}\right) f'\left(\frac{\pi}{4}\right) + \frac{\left(x - \frac{\pi}{4}\right)^2}{2!} f''\left(\frac{\pi}{4}\right) + \dots \quad (1)$$

$$f(x) = \tan^{-1}x \quad f\left(\frac{\pi}{4}\right) = \tan^{-1}\left(\frac{\pi}{4}\right) \quad \left[\tan \frac{\pi}{4} = 1\right]$$

$$f'(x) = \frac{1}{1+x^2} \quad f'\left(\frac{\pi}{4}\right) = \frac{1}{1 + \frac{\pi^2}{16}} = \frac{16}{16 + \pi^2}$$

$$f''(x) = -\frac{1}{(1+x^2)^2} (2x) \quad f''\left(\frac{\pi}{4}\right) = -\frac{2 \cdot \frac{\pi}{4}}{\left(1 + \frac{\pi^2}{16}\right)^2} = -\frac{\pi \cdot 16^2}{2(16 + \pi^2)^2} = -\frac{\pi(128)}{(16 + \pi^2)^2}$$

$$\begin{aligned} f(x) = \tan^{-1}x &= \tan^{-1}\left(\frac{\pi}{4}\right) + \left(x - \frac{\pi}{4}\right) \left(\frac{16}{16 + \pi^2}\right) + \frac{\left(x - \frac{\pi}{4}\right)^2}{2!} \left(\frac{-\pi \cdot 128}{(16 + \pi^2)^2}\right) + \dots \\ &= \tan^{-1}\left(\frac{\pi}{4}\right) + \left(x - \frac{\pi}{4}\right) \left(\frac{16}{16 + \pi^2}\right) - \frac{64\pi}{(16 + \pi^2)^2} \left(x - \frac{\pi}{4}\right)^2 + \dots \end{aligned}$$

find exp. of $\sin x$ in terms of $(x + \frac{\pi}{4})$

$$a = -\frac{\pi}{4}$$

$$\text{but } x = -\frac{\pi}{4} \quad \left(x + \frac{\pi}{4}\right)$$

Expand $\log x$ in powers of $(x-1)$ hence show that

$$\log 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots \quad \checkmark$$

By Taylor's series

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!} f''(a) + \dots$$

$$f(x) = f(1) + (x-1)f'(1) + \frac{(x-1)^2}{2!} f''(1) + \dots \quad (1)$$

$$f(x) = \log x$$

$$f(1) = 0$$

$$f'(x) = \frac{1}{x}$$

$$f'(1) = 1$$

$$f''(x) = -\frac{1}{x^2} = -x^{-2}$$

$$f''(1) = -1$$

$$f'''(x) = -(-2)x^{-3} = 2x^{-3}$$

$$f'''(1) = 2$$

$$f^{(4)}(x) = -6x^{-4}$$

$$f^{(4)}(1) = -6$$

from

$$\log x = 0 + (x-1)(1) + \frac{(x-1)^2}{2!}(-1) + \frac{(x-1)^3}{3!}(2) + \frac{(x-1)^4}{4!}(-6) + \dots$$

$$= (x-1) - \frac{(x-1)^2}{2} + \frac{2!}{6 \cdot 3} (x-1)^3 - \frac{6!}{24 \cdot 4} (x-1)^4 + \dots$$

$$= (x-1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \frac{(x-1)^4}{4} + \dots \quad \checkmark$$

put $x=2$

$$\log 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$$

Apply Taylor's series to find approx. value of
 $f\left(\frac{11}{10}\right)$ where $f(x) = x^3 + 3x^2 + 15x - 10$

$f(x+h) = f(x) + h f'(x) + \frac{h^2}{2!} f''(x) + \dots$

nearest.

$$f\left(\frac{11}{10}\right) = f(1.1)$$

$$\cancel{1.9}$$

$$\cancel{2.01}$$

$$= f(1 + 0.1)$$

$$x=1, h=0.1$$

$$= f(1) + (0.1) f'(1) + \frac{(0.1)^2}{2!} f''(1) + \dots \text{--- (1)}$$

$$f(x) = x^3 + 3x^2 + 15x - 10$$

$$f'(x) = 3x^2 + 5x + 15$$

$$f''(x) = 6x + 5$$

$$f'''(x) = 6$$

$$f^{(4)}(x) = 0$$

from (1)

$$f(1) = 1 + 3 + 15 - 10 = 9$$

$$f'(1) = 3 + 5 + 15 = 24$$

$$f''(1) = 6 + 5 = 11$$

$$f'''(1) = 6$$

$$f(1+0.1) = 9 + (0.1)(24) + \frac{(0.1)^2}{2!}(11) + \frac{(0.1)^3}{3!}(6) + 0$$

$$= 9 + 2.4 + 0.055 + 0.001 = 11.456$$

* Expand $\tan^{-1}(x+h)$ in powers of h & hence find
 $\tan^{-1}(1.003)$ upto 5 places of decimals

given $\pi = 3.141593$



$$f(x+h) = f(x) + h f'(x) + \frac{h^2}{2!} f''(x) + \frac{h^3}{3!} f'''(x) + \dots \quad (1)$$

$$f(x+h) = \tan^{-1}(x+h)$$

$$f(x) = \tan^{-1} x$$

$$f'(x) = \frac{1}{1+x^2}$$

$$\left(\frac{u}{v}\right)' = \frac{vu' - uv'}{v^2}$$

$$\begin{aligned} f'(x) &= -\frac{1}{(1+x^2)^2} (2x) = -\frac{2x}{(1+x^2)^2} \\ f''(x) &= -2 \left[\frac{(1+x^2)^2 (1) - x \cdot 2(1+x^2)(2x)}{(1+x^2)^4} \right] \\ &= -2(1+x^2) \left[\frac{(1+x^2) - 4x^2}{(1+x^2)^3} \right] \\ &= -2 \left[\frac{1-3x^2}{(1+x^2)^3} \right] = 2 \left[\frac{3x^2-1}{(1+x^2)^3} \right] \end{aligned}$$

from (1)

$$\tan^{-1}(x+h) = \tan^{-1} x + h \left(\frac{1}{1+x^2} \right) + \frac{h^2}{2!} \left(\frac{-2x}{(1+x^2)^2} \right) + \frac{h^3}{3!} \left[\frac{2(3x^2-1)}{(1+x^2)^3} \right] + \dots$$

$$\tan^{-1}(1.003) = \tan^{-1}(1+0.003) \quad x=1, h=0.003$$

$$= \tan^{-1}(1) + \frac{0.003}{1+1} - \frac{(0.003)^2}{(1+1)^2} + \frac{(0.003)^3}{3} \left[\frac{3-1}{2^3} \right]$$

$$= \frac{\pi}{4} + \frac{0.003}{2} - \frac{(0.003)^2}{4} + \frac{(0.003)^3}{3 \times 4}$$

$$= \frac{3.141593}{4} + \frac{0.003}{2} - \frac{(0.003)^2}{4} + \frac{(0.003)^3}{12} = 0.786896 \approx 0.78690$$

Using Taylor's series / Thm arrange in powers of x

$$7 + (x+2) + 3(x+2)^2 + (x+2)^4 - (x+2)^5$$

$$f(x+h) = f(h) + x f'(h) + \frac{x^2}{2!} f''(h) + \dots \quad (1)$$

given $f(x+2) = 7 + (x+2) + 3(x+2)^2 + (x+2)^4 - (x+2)^5$
 $h=2$

from (1) $f(x) = 7 + x + 3x^3 + x^4 - x^5$

$$f(x+2) = f(2) + x f'(2) + \frac{x^2}{2!} f''(2) + \dots \quad (2)$$

$$f(x) = 7 + x + 3x^3 + x^4 - x^5 \quad f(2) = 17$$

$$f'(x) = 1 + 9x^2 + 4x^3 - 5x^4 \quad f'(2) = -11$$

$$f''(x) = 18x + 12x^2 - 20x^3 \quad f''(2) = -76$$

$$f'''(x) = 18 + 24x - 60x^2 \quad f'''(2) = -174$$

$$f^{(4)}(x) = 24 - 120x \quad f^{(4)}(2) = -216$$

$$f^{(5)}(x) = -120 \quad f^{(5)}(2) = -120$$

$$f(x+2) = 17 + x(-11) + \frac{x^2}{2!}(-76) + \frac{x^3}{3!}(-174) + \frac{x^4}{4!}(-216) + \frac{x^5}{5!}(-120) + 0$$

$$= 17 - 11x - 38x^2 - 29x^3 - 9x^4 - x^5$$

QW

Find expansion of $\tan(x + \pi/4)$ in ascending powers

x upto
terms in x^4

$$h = \pi/4$$

$$f(x + \pi/4) = \tan(x + \pi/4)$$

$$f(x) = \tan x$$

$$f(x+h) = f(h) + x f'(h) + \frac{x^2}{2!} f''(h) + \frac{x^3}{3!} f'''(h) + \frac{x^4}{4!} f^{(4)}(h) + \dots$$

Maclaurin's Series

$$f(x) = f(0) + x \underline{f'(0)} + \frac{x^2}{2!} \underline{f''(0)} + \frac{x^3}{3!} \underline{f'''(0)} + \dots \quad (1)$$

① $f(x) = e^x$
 $f(x) = e^x$
 $f'(x) = e^x$
 $f''(x) = e^x$

$f(0) = 1$
 $f'(0) = 1$
 $f''(0) = 1$

$e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots$
 $a^x = e^{\log a^x} = e^{(\log a)x}$
 $= 1 + (\log a)x + \frac{(\log a)^2 x^2}{2!} + \dots$

$f(x) = e^{(\log a)x}$
 $f'(x) = e^{(\log a)x} (\log a)$
 $f'(0) = \log a$
 ~~$y_n = e^{ax} a^n$~~

from ① $f(x) = e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \dots$

② $f(x) = \sin x$

$f(x) = \sin x$
 $f'(x) = \cos x$
 $f''(x) = -\sin x$
 $f'''(x) = -\cos x$
 $f^{(4)}(x) = \sin x$
 $f^{(5)}(x) = \cos x$

$f(0) = 0$
 $f'(0) = 1$
 $f''(0) = 0$
 $f'''(0) = -1$
 $f^{(4)}(0) = 0$
 $f^{(5)}(0) = 1$

~~$f(x) = a^x$
 $f'(x) = a^x \log a$
 $f'(0) = \log a$~~

$$f(x) = \sin x = 0 + x + \frac{x^2}{2!}(0) + \frac{x^3}{3!}(1) + \frac{x^4}{4!}(0) + \frac{x^5}{5!}(-1) + \dots$$

$$= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

H.W $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$

PT $\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots$

$f(x) = \cosh x$	$\text{HW} \Rightarrow \sinh x$	$\cosh x = \frac{e^x + e^{-x}}{2}$
$f(x) = \cosh x$	$f(0) = \cosh(0) = 1$	$\sinh x = \frac{e^x - e^{-x}}{2}$
		$\sinh(0) = 0$

$$f'(x) = \sinh x \quad f'(0) = 0$$

$$f''(x) = \cosh x \quad f''(0) = 1$$

$$f'''(x) = \sinh x \quad f'''(0) = 0$$

$$\begin{aligned}
 f(x) &= f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \dots \\
 &= 1 + x(0) + \frac{x^2}{2!} + 0 + \frac{x^4}{4!} + \dots \\
 &= 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots
 \end{aligned}$$

HW p. 7. $\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots$

$$f(x) = \log(1+x)$$

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \dots$$

$$f(x) = \log(1+x) \quad f(0) = 0$$

$$f'(x) = \frac{1}{1+x} \quad f'(0) = 1$$

$$f''(x) = -\frac{1}{(1+x)^2} = -(1+x)^{-2} \Rightarrow f''(0) = -2$$

$$f'''(x) = +2(1+x)^{-3} \Rightarrow f'''(0) = -6$$

$$f^{(4)}(x) = -6(1+x)^{-4} \Rightarrow f^{(4)}(0) = 24$$

$$f^{(5)}(x) = 24(1+x)^{-5} \Rightarrow f^{(5)}(0) = -120$$

$$\begin{aligned} \log(1+x) &= 0 + x(1) + \frac{x^2}{2!}(-2) + \frac{x^3}{3!}(-6) + \frac{x^4}{4!}(24) + \frac{x^5}{5!}(-120) + \dots \\ &= x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \frac{x^6}{6} + \dots \quad \checkmark \end{aligned}$$

$$\text{HW} \rightarrow \log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \frac{x^5}{5} - \dots \quad \checkmark$$

$$\begin{aligned} \textcircled{2} \quad \tanh^{-1} x &= \frac{1}{2} \log\left(\frac{1+x}{1-x}\right) = \frac{1}{2} \left[\log(1+x) - \log(1-x) \right] \\ &= \frac{1}{2} \left[x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \dots - \left(-x - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \frac{x^5}{5} - \dots \right) \right] \\ &= \frac{1}{2} \left[2x + \frac{2x^3}{3} + \frac{2x^5}{5} + \dots \right] = x + \frac{x^3}{3} + \frac{x^5}{5} + \dots \end{aligned}$$

HW

$$\sinh^{-1} x = \log(x + \sqrt{x^2+1}) = x + \frac{1}{2} \frac{x^3}{3} + \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{x^5}{5} + \dots$$

$$\left\{ \begin{aligned} \sinh^{-1} x &= x + \frac{1}{2} \frac{x^3}{3} + \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{x^5}{5} + \dots \\ \csc^{-1} x &= \frac{\pi}{2} - \left[x + \frac{1}{2} \frac{x^3}{3} + \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{x^5}{5} + \dots \right] \\ \tan^{-1} x &= x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots \end{aligned} \right.$$

$$f(x) = (1+x)^m$$

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \dots \quad (1)$$

$$f(x) = (1+x)^m \quad f(0) = 1$$

$$f'(x) = m(1+x)^{m-1} \quad f'(0) = m$$

$$f''(x) = m(m-1)(1+x)^{m-2} \quad f''(0) = m(m-1)$$

$$f'''(x) = m(m-1)(m-2)(1+x)^{m-3} \quad f'''(0) = m(m-1)(m-2)$$

from (1)

$$f(x) = 1 + x m + \frac{x^2}{2!} m(m-1) + \frac{x^3}{3!} m(m-1)(m-2) + \dots \quad (*)$$

Note If m is +ve int. then derivative is vanishing
 $\therefore$ we get finite series.

If $m = -1$

from

$$(1+x)^{-1} = \frac{1}{1+x} = 1 - x + \frac{x^2}{2!} (2) + \frac{x^3}{3!} (1)(-2)(-3) + \dots$$

$$= 1 - x + x^2 - x^3 + x^4 \dots \quad \checkmark$$

$$(1-x)^{-1} = \frac{1}{1-x} = 1 + x + x^2 + x^3 + x^4 \dots \quad \checkmark$$